STUDENT MARK ANALYZER SYSTEM

A Self-Developed Mini Project Using Python and SQLite Database

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Submitted By

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Project Details

Project Title: Student Mark Analyzer System
Type: Console-Based Application
Domain: Educational Data Management System
Programming Language: Python
Database Used: SQLite
Libraries Used: sqlite3, pandas
Development Type: Individual Project (Self Developed)

Project Report on

STUDENT MARK ANALYZER SYSTEM

(Using Python and SQLite Database)

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1. Introduction

The Student Mark Analyzer System is a software application developed using the Python programming language and SQLite database management system. The primary purpose of this system is to simplify and automate the process of managing student academic records in an educational environment.

In traditional systems, student marks are recorded manually, and calculations such as total marks, average marks, and grade assignment are performed manually by teachers or administrators. This process is not only time-consuming but also prone to human errors. To overcome these challenges, this system provides an automated solution that efficiently handles student data and performs all required calculations instantly.

The system also includes a secure login mechanism for administrators. This ensures that only authorized users can access and modify the student database. Once logged in, the admin can add new student records, view all students, and search for specific students using their names.

Overall, this project demonstrates how Python and SQLite can be combined to create a simple yet powerful academic management system.

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2. Objective of the Project

The main objective of this project is to design and develop a simple, efficient, and automated student record management system that reduces manual effort and improves accuracy in academic data handling.

This system aims to provide a structured way of storing student information and performing automatic calculations related to academic performance.

The specific objectives of this project are as follows:

- To store and manage student academic records in a structured database format
- To automate the calculation of total marks and average marks
- To generate grades automatically based on student performance
- To provide a search facility for quick retrieval of student records
- To implement a secure login system for data protection
- To gain practical knowledge of database management using SQL and Python integration
- To understand real-world application development using programming concepts

This project helps in bridging the gap between theoretical knowledge and practical implementation of software systems.

3. Technologies Used

This project is developed using modern yet simple technologies that are suitable for beginner-level to intermediate-level software development.

Programming Language

Python is used as the core programming language due to its simplicity, readability, and powerful libraries.

Database

SQLite is used as the database system. It is a lightweight, file-based database that does not require any external server setup, making it ideal for small applications.

Libraries Used

The project uses the following Python libraries:

• sqlite3 – This library is used to create and manage the database, execute SQL queries, and handle data storage operations.
• pandas – This library is used for data analysis, tabular representation, and performing calculations such as total marks and average marks efficiently.

These technologies together make the system efficient, lightweight, and easy to execute on any system.

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4. System Design

The Student Mark Analyzer System follows a simple modular design approach. The system is divided into different components, each responsible for a specific functionality such as user authentication, data entry, data retrieval, and analysis.

The system architecture is based on three main layers: user interface layer, application logic layer, and database layer. The user interacts with the system through a command-line interface. The application logic processes the input and performs required operations such as calculations and database queries. The database layer stores all the data permanently.

4.1 Database Design

The system uses a database named student_system.db, which contains two main tables.

Admin Table

The admin table is used to store login credentials of authorized users. It ensures system security by allowing only valid users to access the system.

Field	Type	Description
id	INTEGER	Primary Key
username	TEXT	Admin username
password	TEXT	Admin password

This table plays a crucial role in authentication and system protection.

Students Table

The students table is used to store academic information of students, including their marks in different subjects.

Field	Type	Description
id	INTEGER	Primary Key
name	TEXT	Student name
math	INTEGER	Math marks
science	INTEGER	Science marks
english	INTEGER	English marks

This structured format helps in easy retrieval and analysis of student data.

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5. Working of the System

The working of the system is divided into multiple steps, ensuring smooth execution and user-friendly interaction.

Step 1: Login System

In the first step, the administrator is required to enter a username and password. The system verifies these credentials by matching them with the records stored in the admin table. If the credentials are correct, access is granted; otherwise, the system denies entry.

Step 2: Add Student

After successful login, the admin can add student details. The system takes input such as student name and marks in three subjects. These details are then stored permanently in the SQLite database.

Step 3: View Students

This module retrieves all student records from the database and displays them in a structured format. The system then performs calculations such as total marks, average marks, and grade assignment automatically using predefined logic.

Step 4: Search Student

This module allows the user to search for a student by name. It uses SQL query matching to find relevant records and displays complete details of the student if found.

The system ensures smooth flow between all modules through a menu-driven interface.

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6. Grade Calculation Logic

The grading system is based on the average marks obtained by the student. The system calculates the average by dividing the total marks by the number of subjects.

Average Marks	**Grade**
90 – 100	A+
75 – 89	A
60 – 74	B
Below 60	C

This grading logic helps in evaluating student performance in a standardized manner and provides clear academic classification.

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7. Features of the Project

The Student Mark Analyzer System includes several important features that make it useful and efficient for academic environments.

- Secure login system for admin authentication
- Ability to add and store student records in a database
- Automatic calculation of total marks and average marks
- Grade assignment based on performance
- Search functionality for quick student retrieval
- Simple and user-friendly menu-driven interface
- Lightweight system that runs without internet or server dependency

These features make the system practical for small-scale educational institutions.

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8. Advantages

The system provides multiple benefits in terms of efficiency, accuracy, and usability.

• Reduces manual workload of teachers and administrators
• Eliminates calculation errors in academic records
• Provides fast access to student data
• Ensures structured and organized data storage
• Works offline without requiring internet connection
• Easy to use and understand for beginners

Overall, the system improves the efficiency of academic record management.

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9. Limitations

Despite its advantages, the system has some limitations that can be improved in future versions.

• It does not have a graphical user interface (GUI)
• It is limited to small-scale datasets
• It does not support online or cloud-based access
• Passwords are stored in plain text, which is not secure
• It lacks advanced data visualization features

These limitations highlight areas for future enhancement.

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10. Future Enhancements

The system can be improved and expanded in several ways to make it more advanced and industry-ready.

• Development of a web-based application using Flask or Django
• Addition of graphical dashboard with charts and graphs
• Implementation of student ranking system
• Export of reports in PDF and Excel formats
• Integration with cloud-based databases
• Secure password encryption system

These enhancements will make the system more powerful and suitable for real-world applications.

11. Conclusion

The Student Mark Analyzer System is a simple yet effective application designed to manage and analyze student academic records efficiently. It successfully automates important tasks such as calculation of total marks, average marks, and grade assignment.

The project demonstrates the practical use of Python programming, SQLite database management, and basic data analysis using Pandas. It helps in reducing manual work, improving accuracy, and organizing student data in a structured format.

This project provides a strong foundation for understanding real-world software development concepts and can be further enhanced into a more advanced academic management system in the future.

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12. Developed By

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Date: 27 July 2025
